Windows ISO Toolkit (All-in-One)

Three tools in one repo for debloating, customizing, and repairing Windows ISOs — before AND after installation.

Toolkit Ecosystem

Tool	Type	What It Does	When to Use
isoDebloaterScript.ps1	ISO debloater	Strips bloat from a Windows ISO before installation. Removes AppX packages, features, Edge, AI, OneDrive, services, telemetry, and builds a debloated bootable ISO.	**Before installing Windows** — create a clean ISO to install from scratch.
rintechtoolkit\isoToolkit.ps1	ISO modding / repair toolkit	Menu-driven toolkit for integrating updates, drivers, .NET, registry tweaks, OEM branding, wallpapers, autounattend.xml, browser data import, WIM operations, WinRE recovery, and repairing/restoring features from a donor ISO.	**After debloating** — fine-tune the ISO with customizations, restore removed components, or repair corrupted images.
BakuretsuClean	Live OS debloater	Debloats an already-installed Windows system. Removes AppX packages, disables services, blocks telemetry, applies privacy tweaks. Includes revert support.	**After installing Windows** — debloat a live system without touching the ISO.

Recommended Workflow

```
1. isoDebloaterScript.ps1  →  Strip bloat from stock ISO, build debloated ISO
2. rintechtoolkit\          →  Customize/repair the debloated ISO (drivers, wallpaper, WinRE recovery, etc.)
3. Install the debloated ISO →  Test in VM or bare metal
4. BakuretsuClean\          →  (Optional) Apply additional live debloat after installation
5. Cleanup.bat              →  Clear all temp files, logs, backups, and output ISOs for a fresh start
```

Quick Start

ISO Debloater:

1. **Run PowerShell as Administrator**
2. Execute: `.\isoDebloaterScript.ps1`
3. Select your Windows ISO when prompted
4. Answer the debloat options (press Enter for defaults)
5. Wait for the debloated ISO to be created

ISO Toolkit (modding / repair):

1. Run: `.\rintechtoolkit\isoToolkit.ps1`
2. Select `1. Source` → load an ISO
3. Use `Integrate`, `Customize`, `Tools` menus to modify or repair the image
4. `5. Save changes & unmount` → `6. Build ISO`

Live OS Debloater:

1. Run: `BakuretsuClean\RunDebloater.bat` as Administrator

Cleanup: Run `Cleanup.bat` to wipe all work files, logs, temp dirs, backup ISOs, and output ISOs.

Minimum requirements: Windows 10/11 host, 20GB free disk space, Administrator privileges.

Debloat Tiers

The ISO debloater has **three tiers** of aggressiveness. Each tier includes everything from the one below it.

Tier 1: Standard (`-SafeMode` or select options individually)

The baseline. Removes AppX bloatware, optional features, OneDrive, Edge, AI components, and applies privacy/performance tweaks. **Safe for any Windows edition.**

Category	What's Removed
AppX packages (85+)	Bing (News, Weather, Search, Translator), Xbox (App, GameOverlay, GamingOverlay, SpeechToText, TCUI), Skype, Teams, OneNote, Clipchamp, 3D Viewer, Maps, Solitaire, Mixed Reality, People, Camera, Photos, Paint3D, Wallet, GamingApp, GetHelp, GetStarted, FeedbackHub, MicrosoftFamily, OfficeHub, Outlook, PowerAutomate, QuickAssist,

Category	What's Removed
	StickyNotes, ToDos, Zune (Music, Video), Alarms, DevHome, CrossDevice, CommunicationsApps, PeopleExperienceHost, WebExperience, Calling, CapturePicker, NarratorQuickStart, ParentalControls, Print3D, SecureAssessmentBrowser, XGpuEjectDialog, Advertising, Journal, ECApp, AV1/HEVC/HEIF/VP9/MPEG2/Raw/Dolby codec extensions, ScreenSketch, WebMediaExtensions
Capabilities (15+)	Internet Explorer, WordPad, PowerShell ISE, Media Player, Steps Recorder, Quick Assist, Print 3D, Wireless Display, Print Management Console, handwriting/OCR/speech/text-to-speech for detected language
Windows packages (10+)	Internet Explorer, WordPad, MediaPlayer, TabletPCMath, StepsRecorder, QuickAssist, PowerShell ISE, printing PMCPPC/WFS, XPS Viewer, ADAM client
OneDrive	Setup.exe (SysWOW64 + System32), shortcuts, WinSxS components, appdata folders, registry run keys
Edge	DISM Remove-Edge, Stable + DevTools AppX, EdgeUpdate registry/service, Program Files, scheduled tasks, WinSxS components, WebView2 (Win10 only — kept on Win11 for OOBE)
AI / Copilot	Copilot, Recall, AIX, CoreAI, GameAssist, WritingAssistant, OfficeActionsServer AppX packages + 30+ registry policies blocking AI in Search, Edge, Paint, Notepad, Office, and system-wide. Hidden CBS packages unhidden and removed.
Widgets	WebExperience + Widgets AppX packages, News & Interests policies, scheduled tasks, WinSxS components
Services disabled (40+)	DiagTrack, dmwappushsvc, WSearch, SysMain, MapsBroker, lfsvc, PhoneSvc, MessagingService, PimIndexMaintenanceSvc, BcastDVRUserService, XblAuthManager, XblGameSave, XboxNetApiSvc, XboxGipSvc, WalletService, PrintNotify, LicenseManager, WbioSrvc, FrameServer, CaptureService, SensorService, SensorDataService, SensrSvc, TabletInputService, TrkWks, FontCache, DusmSvc, WpnService, StiSvc, wisvc, RetailDemo, tzautoupdate, W32Time, wcncsvc,

Category	What's Removed
	WerSvc, WiaRpc
Registry / Privacy	Sponsored apps, telemetry (AllowTelemetry=0), mouse acceleration, Meet Now, ads & suggestions, BitLocker auto-encryption, GameDVR/GameBar, Cortana, consumer features, location tracking, Find My Device, Windows Tips, background apps, web search, error reporting, delivery optimization, advertising ID, clipboard history, activity feed, timeline, setting sync, Windows Ink, lock screen suggestions
Performance	Disable visual effects, animations, hibernation. Optimize CPU priority (Win32PrioritySeparation=38), large system cache, disable paging executive, network throttling, ICMP redirects, Game Mode

Approximate footprint: ~18–22 GB free on a 126 GB disk (varies by edition).

Tier 2: Micro (`-MicroMode yes`)

Everything in Tier 1, plus aggressive stripping. **Comparable to Micro 10/11 builds.** Breaks OOBE (use `-OOBEBypass yes`). Auto-enables: ExtremeDebloat, DefenderRemove, WinUpdateDisable, HyperVRemove, TaskCleanup, WinSxSCleanup, TPMBypass.

Category	What's Added Beyond Tier 1
Extreme services	wlidsvc (MS Account), UnistoreSvc, UserDataSvc, OneSyncSvc, Spooler (printing), LanmanServer (file sharing), LanmanWorkstation, fdPHost, FDResPub, SSDP/UPnP, SCardSvr, ScDeviceEnum, SmsRouter, BTAGService (Bluetooth audio), bthserv (Bluetooth), Fax, hidserv (keyboard/mouse special keys)
Additional AppX	DesktopAppInstaller (winget), StorePurchaseApp, WebMediaExtensions, AssignedAccessLockApp, SecureAssessmentBrowser
Additional capabilities	Hello.Face, OneCoreUAP.OneSync, OpenSSH.Client, Xps.Viewer, Print.Fax.Scan, MathRecognizer, ALL language handwriting/OCR/speech/TTS for all languages
Additional packages	Hello-Face, Print-Fax-Scan-Feature, VBSCRIPT, Legacy WOW64 compatibility, Notepad (Win10), MSPaint (Win10),

Category	What's Added Beyond Tier 1
	MediaPlayer
Defender (aggressive)	11 services disabled (including WdBoot/WdFilter/WdDevFlt kernel drivers + Security Center), 40+ registry policy keys, files stripped from Program Files + ProgramData + System32 + drivers, 7 WinSxS patterns, scheduled tasks nuked, 8 post-install RunOnce cleanup commands
WinRE	Winre.wim deleted, Recovery directory removed, WinRE registry disabled
Windows Update	All WU services disabled, WU scheduled tasks deleted, WU/waasmedic/updateorchestrator WinSxS components removed, WU orchestrator registry nuked, DoNotConnectToWindowsUpdateInternetLocations policy set
Fonts	All non-essential fonts removed — keeps only Segoe UI, Arial, Times New Roman, Courier New, Marlett, Symbol, Wingdings, Tahoma, Verdana, CJK system fonts
WinSxS (deep)	50 patterns: winre, recovery, bitlocker, security, defender, speech, hyperv, virtual, containers, sandbox, WSL, print, fax, scanner, xps, hello, biometrics, narrator, accessibility, magnifier, screen, camera, tablet, touch, pen, ink, 3D, holographic, terminal, powershell, netfx, IIS, workfolders, branchcache, directaccess, RAS, VPN, NDIS, mobile, telephony, ADAM, LDAP, MSMQ, multipoint, RDMA, storage replica, tiering, DFS, failover
Micro registry	Store blocking, Mail/Calendar disable, live tiles off, push notifications off, all background apps force-deny, lock screen off, task view off, people bar off, news/feeds off, widgets off, nearby sharing off, game DVR off, Cortana force-disable, find my device off, sync force-disable, ink workspace off, clipboard history/cloud off, activity feed off

Approximate footprint: ~8–12 GB free on a 126 GB disk.

Tier 3: Ultra Micro (`-UltraMicroMode yes`)

Everything in Tier 2, plus extreme disk space recovery. **Closes the ~10 GB gap with ultra-stripped Micro 10 builds (122 GB free on 126 GB disk).** Auto-enables MicroMode +

OOBEBypass + everything.

Category	What's Added Beyond Tier 2
Servicing stack backups	Entire `WinSxS\Backup\` directory stripped (~1–3 GB)
Servicing manifests	60+ bloat component manifests deleted from `WinSxS\Manifests\` — defender, edge, xbox, skype, hyperv, print, fax, scanner, and more (~500 MB–1 GB)
Driver store	Non-critical drivers stripped: printer, scanner, fax, modem, bluetooth, wifi, wwan, sensor, camera, biometric, smartcard, NFC, touch, pen, tablet, audio, media, DRM, display, battery, thermal (~500 MB–2 GB)
NGEN cache	Pre-compiled .NET assemblies removed from `assembly\NativeImages*` and `Microsoft.NET\assembly\GAC_MSIL`; .NET will JIT on first run (~300–500 MB)
Language MUI files	All `xx-XX` directories in `System32` and `SysWOW64` removed except the detected language (~300 MB–1 GB)
Visual resources	All wallpapers, themes, cursors, 4K wallpapers, screen images deleted (~100–200 MB)
Migration data	`System32\migration`, `migwiz`, `oobe\migrate`, `Sysprep` files stripped (~100–300 MB)
Reserved storage	All ReserveManager registry keys zeroed; RunOnce fires `compact /compactos:always` on first boot (~7 GB live savings)
Ultra WinSxS	50 additional broad patterns: servicing, winsat, migration, upgrade, compatibility, driver, remotedesktop, DVD, bluray, media, location, wallet, parentalcontrols, family, easeofaccess, windowstogo, spelling, dictation, networking, SNMP, telnet, TFTP, RIP, LPD, LPR, IPC, RPC (~500 MB–2 GB)
Ultra registry + RunOnce	System restore disable, prefetch/superfetch disable, pagefile disable, dism /resetbase run-once, WinSxS Backup/ManifestCache strip on live system
Additional AppX	Cortana, Search, Apprep.ChxApp, ECApp
Additional packages	IE, StepsRecorder, QuickAssist, PowerShell ISE, printing PMCPPC/WFS, XPS Viewer, ADAM client

Approximate footprint: ~~2–4 GB used on a 126 GB disk (~~122 GB free).

Which Tier Should You Use?

Tier	Command	Use Case
Standard	`-SafeMode` or pick options manually	Safe for any edition (Home/Pro/Enterprise/LTSC). No OOBE issues.
Micro	`-MicroMode yes`	Maximum stripping with working desktop. Combine with `-OOBEBypass yes` to skip the broken OOBE.
Ultra Micro	`-UltraMicroMode yes`	Closest to Micro 10/11 builds. Maximum free disk space. OOBEBypass auto-enabled.

Cleanup

After one or more debloat/modding sessions, leftover files accumulate:

Leftover	Source
`C:\WIDTemp\`	isoDebloaterScript.ps1 work directory
`C:\ISOToolkit\`	isoToolkit.ps1 work directory
`ISOBackup\`	Original ISO backups (~6 GB each)
`ErrorLogs\`	Timestamped error logs from all tools
`ADKDownload\`	Cached oscdimg / ADK downloads
`*.iso` (output ISOs)	Debloated/custom output ISOs
`*.log`, `*.txt`	Script/transcript logs
`%TEMP%\toolkit_*.reg`	Temp registry import files

Run **`Cleanup.bat`** (as Administrator) from the toolkit folder to clear everything. It prompts before removing each ISO so you can keep ones you want. After cleanup, the toolkit folder returns to its initial state — ready for a fresh run.

Virtual Machine Troubleshooting

> **IMPORTANT:** These issues are caused by **VM configuration quirks** and the **natural side effects of debloating —** they are **NOT defects in your debloated ISO.** Do not file a bug report for these.

1. Endless "Install Now" Loop (Hyper-V Gen 1)

Symptom: After installation reaches 100% and reboots, the VM drops back into the initial setup screen endlessly.

Root Cause: Hyper-V Generation 1 VMs boot from the attached virtual DVD drive before the virtual hard disk on every restart.

Fix: Power off the VM completely. Go to **VM Settings** → **DVD Drive** → set **Media** to **"None"** (unmount the ISO). Apply and start the VM cold. Alternatively, do not press any key when *"Press any key to boot from CD..."* appears.

2. OOBE Stuck at Region / "Just a Moment" Loop

Symptom: The system hangs on the blue "Just a moment..." screen or loops endlessly at the region/keyboard selection screen.

Root Cause: Debloating strips the consumer telemetry and online account frameworks that the OOBE wizard expects. The interactive OOBE panics when its online hooks are absent.

Fix:

1. At the stuck screen, press **Shift + F10** (or **Fn + Shift + F10** on laptops) to open Command Prompt.
2. Enable the built-in Administrator account:

   ```
   net user administrator /active:yes
   ```

3. Kill the stuck OOBE process and relaunch it properly:

   ```
   cd C:\Windows\System32\oobe
   msoobe
   ```

4. The OOBE wizard will restart. Go through the screens normally — it should now proceed past the region section.
5. **If testing in Hyper-V:** After completing the OOBE, you **must restart the virtual machine** (Action → Reset or Ctrl+Alt+End → restart). Hyper-V does not properly transition from the OOBE environment to the desktop without a cold restart.
6. On first login, you can set up your own user account and disable the built-in Administrator if desired.

3. Automatic OOBE Bypass (Built Into the Debloater)

Instead of fixing OOBE loops manually, you can have the debloater **inject an automatic bypass** directly into the ISO. When enabled (`-OOBEBypass yes` or answer Yes at the prompt), the debloater does two things:

1. **Injects `SetupComplete.cmd`** into the ISO (`\Windows\Setup\Scripts\`). This script runs at the very end of Windows Setup and:
   - Creates a local Administrator account (`Admin`, no password)
   - Configures auto-logon so the system boots straight to desktop

- Force-sets all OOBE bypass registry keys (`OOBEInProgress=0`, `SetupType=0`, `BypassNRO=1`, etc.)
- Disables the OOBE scheduled tasks that would otherwise fire

2. **Pre-seeds offline registry keys** in the mounted image so the bypass is baked in before the first boot even happens.

Result: After installation completes, the system reboots once and lands directly on the desktop as `Admin` — no OOBE screens, no region selection, no account creation, no "Just a moment" loop. Zero user interaction needed after install.

How to use:

```
.\isoDebloaterScript.ps1 -OOBEBypass yes
```

Or choose "Yes" at the interactive prompt: *"Automatically bypass OOBE after install?"*

Security note: The auto-created `Admin` account has no password. Set one on first login or disable the account and create your own.

What Happens During Execution

Step	**Description**	**Time**
ISO selection & backup	Copies original ISO to `ISOBackup\` folder	~1-2 min
Mount & extract	Copies ISO contents to temp working directory	~2-5 min
Package removal	Removes AppX packages via DISM	~3-8 min
Edge removal	Strips Edge browser, WebView, EdgeUpdate	~1-3 min
AI removal	Removes Copilot, Recall, AI components	~1-3 min
Registry tweaks	Privacy, performance, telemetry policies	~1-2 min
Service disabling	Unnecessary services disabled	<1 min
Widgets removal	Strips Widgets, News, WebExperience	~30 sec
Hyper-V removal (optional)	Removes virtualization platform	~1-2 min
Defender removal (optional)	Aggressive Defender strip	~1-2 min
Scheduled tasks (optional)	Disables telemetry/bloat tasks	~1 min
WinSxS cleanup (optional)	Conservative bloat component cleanup	~2-5 min
Micro Mode stripping	WinRE, fonts, WU, deep	~5-10 min

Step	Description	Time
(optional)	WinSxS, Micro registry	
Ultra Micro stripping (optional)	Servicing backups, driver store, NGEN, MUI, manifests, CompactOS	~10-20 min
OOBE Bypass injection (optional)	SetupComplete.cmd + offline reg keys	<1 min
Privacy tweaks	Cortana, location, activity, ads policies	<1 min
Image cleanup	DISM component cleanup & compression	~10-20 min
Health check & repair	Checks & repairs component store	~2-5 min
Export WIM	Compresses modified image	~5-20 min
ISO creation	Builds bootable ISO with oscdimg	~2-5 min
Total (Standard)		**~25-50 min**
Total (Micro)		**~35-65 min**
Total (Ultra Micro)		**~50-90 min**

Command-Line Automation

```
# Fully automated Standard debloat
.\isoDebloaterScript.ps1 -noPrompt -isoPath "C:\Win11.iso" -winEdition "Windows
11 Pro" -outputISO "Win11Lite"

# Micro mode — maximum stripping, OOBE bypass auto-injected
.\isoDebloaterScript.ps1 -MicroMode yes -OOBEBypass yes -isoPath "C:\Win11.iso"
-winEdition "Windows 11 Pro" -outputISO "Win11Micro"

# Ultra Micro mode — extreme disk savings, everything auto-enabled
.\isoDebloaterScript.ps1 -UltraMicroMode yes -isoPath "C:\Win11.iso" -winEdition
"Windows 11 Pro" -outputISO "Win11UltraMicro"

# Safe mode — remove only apps, keep everything else
.\isoDebloaterScript.ps1 -SafeMode

# Custom selection
.\isoDebloaterScript.ps1 -AppxRemove yes -CapabilitiesRemove yes -OnedriveRemove
yes -EDGERemove yes -AIRemove yes -DefenderRemove yes -OOBEBypass yes

# LTSC ISO — skip removals, apply only universal optimizations
.\isoDebloaterScript.ps1 -AppxRemove no -CapabilitiesRemove no -OnedriveRemove
no -EDGERemove no -AIRemove no -WidgetsRemove no -DefenderRemove no
-ServicesDisable yes -PerformanceTweaks yes -TPMBypass yes -ESDConvert yes
```

All Parameters

Parameter	Values	Default	Description
`-noPrompt`	(switch)	off	Fully automated — requires `-isoPath`, `-winEdition`, `-outputISO`
`-isoPath`	path string	prompted	Path to source Windows ISO
`-winEdition`	edition name	prompted	Edition name to mount (e.g. "Windows 11 Pro")
`-outputISO`	filename	prompted	Output ISO filename (no extension)
`-SafeMode`	(switch)	off	AppX removal only, safe for any edition
`-AppxRemove`	yes/no	yes	Remove 85+ bloatware AppX packages
`-CapabilitiesRemove`	yes/no	yes	Remove 15+ optional Windows features
`-OnedriveRemove`	yes/no	yes	Completely remove OneDrive
`-EDGERemove`	yes/no	yes	Remove Microsoft Edge
`-AIRemove`	yes/no	yes	Remove Copilot, Recall, all AI components
`-TPMBypass`	yes/no	no	Bypass TPM 2.0, Secure Boot, CPU checks
`-UserFoldersEnable`	yes/no	yes	Restore Desktop/Documents folders on Win11
`-DriverIntegrate`	yes/no	no	Integrate Intel RST/VMD drivers
`-ESDConvert`	yes/no	no	Compress ISO with recovery (ESD) compression
`-useOscdimg`	yes/no	yes	Use oscdimg.exe for reliable ISO build
`-useDISM`	yes/no	no	Force DISM CLI over PowerShell cmdlets
`-DefenderRemove`	yes/no	no	**Aggressive** — strips Defender completely (files, drivers, WinSxS,

Parameter	Values	Default	Description
			Security Center)
`-WidgetsRemove`	yes/no	yes	Remove Widgets and WebExperience
`-WinUpdateDisable`	yes/no	no	Disable Windows Update services + policies
`-HyperVRemove`	yes/no	no	Remove Hyper-V virtualization platform
`-ExtremeDebloat`	yes/no	no	Disable print spooler, Bluetooth, LAN sharing, etc.
`-**MicroMode**`	yes/no	no	**Tier 2** — WinRE, fonts, deep WinSxS, aggressive Defender, WU strip
`-**UltraMicroMode**`	yes/no	no	**Tier 3** — servicing backups, driver store, NGEN, MUI, manifests, CompactOS
`-ServicesDisable`	yes/no	yes	Disable 40+ telemetry/bloat services
`-PerformanceTweaks`	yes/no	yes	CPU, memory, network optimizations
`-TaskCleanup`	yes/no	no	Disable 45+ telemetry scheduled tasks
`-WinSxSCleanup`	yes/no	no	Conservative WinSxS bloat directory removal
`-UseAutounattend`	yes/no	no	Use autounattend.xml for unattended setup (CAUTION)
`-**OOBEBypass**`	yes/no	no	Inject SetupComplete.cmd — auto-creates Admin account, skips OOBE entirely

---

Health Check & Repair

After all modifications but before ISO creation, the script runs a comprehensive health check:

1. **Component store scan** — `dism /CheckHealth` on the mounted image. Parses DISM output to detect corruption.

2. **Automatic repair** — If issues found, runs `dism /RestoreHealth` using:
    - **Backup ISO** as source (if available — the script creates a backup at `ISOBackup\original_backup.iso`)
    - Falls back to Windows Update / built-in repair if no backup
    - Shows explicit BEFORE/AFTER status: `[REPAIRED]` or `[STILL DAMAGED]`
3. **Critical file verification** — Checks 12 critical files exist (explorer.exe, registry hives, boot files, setup.exe)
4. **Registry hive validation** — Loads each hive to verify they're not corrupted
5. **Health summary** — Reports passed checks vs issues found

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Error Logging

Every run creates a timestamped folder in `ErrorLogs\`:

- `errors_YYYY-MM-DD_HH-mm-ss.log` — All errors, warnings, and skips
- `summary_YYYY-MM-DD_HH-mm-ss.txt` — Formatted summary with counts

Errors are categorized by source (DISM, AppX, Mount, Export, Health, etc.) for easy diagnosis.

Skip / Stuck Recovery

- **Press `S`** during any looping operation (package removal, service disabling, scheduled tasks, WinSxS cleanup) to skip the remaining items in that section.
- For DISM operations, the script has a 2-retry timeout mechanism. If a DISM command hangs, it auto-skips after the timeout.
- If the script crashes mid-run leaving a mounted WIM: run `dism /unmount-image /mountdir:"C:\WIDTemp\mountdir\installWIM" /discard` before re-running.

Which ISO Should You Use?

Any Windows edition can be debloated safely. Earlier assumptions that Home/Pro editions were "risky" were based on misdiagnosed VM issues — the actual root cause of installation loops was **incorrect Hyper-V settings and poor OOBE management** (see VM Troubleshooting above), not ISO corruption or edition incompatibility.

ISO Edition	Debloat Impact	Notes
Windows 11/10 Home	Full	Works with all tiers. Use `-OOBEBypass yes` for Micro/Ultra Micro tiers since

ISO Edition	Debloat Impact	Notes
		the stripped online account hooks cause OOBE to stall.
Windows 11/10 Pro / Pro for Workstations	Full	Works with all tiers. Identical behavior to Home for debloat purposes.
Windows Enterprise	Full	Fewer consumer integrations to begin with — many removals will produce "not found" skips (harmless).
Windows LTSC (IoT/Enterprise LTSC)	Light	Already heavily debloated by Microsoft. Most removal targets don't exist in LTSC — expect many "not found" errors (harmless). Only TPM bypass, driver integration, folder shortcuts, compression, performance tweaks, and service disabling are useful.

Key Realization

The installation problems previously attributed to "corrupted ISOs" or "edition incompatibility" were actually caused by:

1. **Hyper-V Generation 1 boot order** — the VM boots the ISO instead of the hard disk on every restart, looping back to "Install Now" (Section 1 of VM Troubleshooting).
2. **Stripped OOBE hooks** — Micro/Ultra Micro tiers remove the telemetry and online account frameworks the OOBE wizard expects, causing it to hang at "Just a moment..." (Section 2 of VM Troubleshooting).

Both are easily fixed — unmount the ISO after install, or use `-OOBEBypass yes` to skip the wizard entirely. The debloated ISO itself is fine.

For Home/Pro ISOs — Use the Tiers

Home and Pro editions respond identically to debloating. Choose the tier that matches your needs:

Tier	Command	Result
Standard	`-SafeMode` or pick options	Clean ISO, OOBE works normally. ~18 GB free.
Micro	`-MicroMode yes` `-OOBEBypass yes`	Heavily stripped, OOBE skipped, auto-logon. ~8-12 GB free.
Ultra Micro	`-UltraMicroMode yes`	Maximum stripping, everything auto-enabled. ~122 GB free.

Or if you prefer to debloat a live system instead of the ISO:

```
BakuretsuClean — live-OS debloater (included in the repo)
Run: BakuretsuClean\RunDebloater.bat as Administrator
Features: AppX removal, service disabling, telemetry blocking, privacy tweaks,
revert support
```

LTSC Guidance

LTSC (Long-Term Servicing Channel) editions are Microsoft's own debloated Windows builds. They ship without Store apps, Widgets, Copilot, AI, Xbox, Bing, and most consumer features. Running aggressive removal options on LTSC will produce **"not found" / "failed" errors** — this is normal because the script tries to remove components LTSC never had.

Recommended: skip these on LTSC

```
-AppxRemove no -CapabilitiesRemove no -OnedriveRemove no -EDGERemove no
-AIRemove no -WidgetsRemove no -DefenderRemove no -TaskCleanup no -WinSxSCleanup
no
```

Still useful on LTSC:

```
-TPMBypass yes -UserFoldersEnable yes -DriverIntegrate yes -ESDConvert yes
-useOscdimg yes -PerformanceTweaks yes -ServicesDisable yes
```

Tips

- **Understand the tiers** — Standard is safe for all editions. Micro breaks OOBE but produces a working desktop (use `-OOBEBypass yes`). Ultra Micro maximizes free space but removes more components.
- **LTSC ISOs are already debloated** — errors during debloating are normal and harmless. Skip most removal options.
- **Always test in a VM first** — before deploying a debloated ISO to real hardware, install it in Hyper-V or VMware to verify it boots and runs.
- **Use a fast SSD** for the working directory — DISM operations are disk-intensive.
- **Disable real-time AV** temporarily — some AV products interfere with DISM mounting operations.
- **The backup ISO** at `ISOBackup\original_backup.iso` can be deleted after confirming the debloated ISO works.
- **Check the error logs** at `ErrorLogs\` if the debloated ISO has installation issues.
- **Use the rintechtoolkit after debloating** — to add drivers, customize wallpaper, set OEM branding, create autounattend.xml, or recover removed components (WinRE, features) from a donor ISO.

Tested Versions

- **Windows 10**: 22H2 (Build 19045)
- **Windows 11**: 24H2 (Build 26100)

Credits

- Created by shirayukimikoto and fujiwarasayo
- Inspired by MSMG Toolkit, KernelOS, and Windows X-Lite stripped builds
- tiny11builder for inspiration
- Winaero for registry optimization techniques
- RemoveWindowsAI for AI removal approach

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